control. In an automatic star-delta starter, time delay relay is used (TDR) through which star to delta connections take place automatically with some pre-fixed time delay. The delay time of the TDR is fixed, keeping in view the starting time of the motor.

Advantages

i. Inexpensive.
ii. Starting current reduced to one-third of direct online starting current.
iii. Easy to operate.

Disadvantages

i. Low starting torque.
ii. Six terminal motor required. (delta connected)
iii. It requires 2 sets of cables from starter to motor.

iii. Auto-transformer starter

An Auto transformer starter consists of an auto transformer and a switch as shown in figure 8.5.

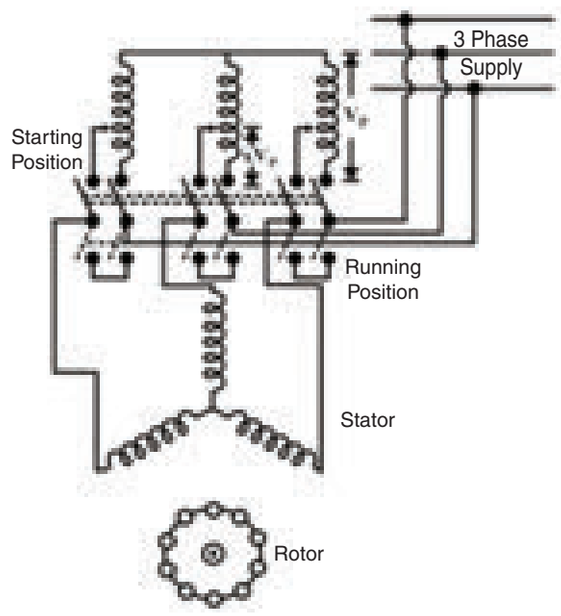


Fig. 8.5 Auto transformer starter

When the switch 'S' is put in start position, a reduced voltage is applied across the motor terminals. When the motor picks up speed, say to 80 percent of its normal speed, the switch is put to run position. Then the auto transformer is cut off the circuit and full rated voltage gets applied across the motor terminals.

The circuit diagram in figure 8.5 is a manual auto-transformer starter. This can be made push button operated automatic controlled starter so that the contact switches over from start to run position as the motor speed picks up of 80% of its speed. Over load protection relay has not been shown in figure 8.5.

The switch 'S' is an air break type for small motors. There are more than one tapping to enable the user to select any suitable starting voltage depending upon the conditions.

iv. Rotor resistance starter

The easiest method of starting slip ring induction motor is to connect some extra resistance in the rotor circuit as shown in figure 8.6.

Connection of extra resistance in the rotor circuit decreases the starting current and at the same time increases the starting torque.

As the motor starts rotating the extra resistance is cut off. When the motor attains rated speed, the resistance is fully cut off and slip ring terminals are short circuited. The motor now operates on its own characteristic which gives rise to maximum torque at a low slip.

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